

MEDIATEK

MT8676 Android General Introduction

Version: 1.1
Release date: 2024-11-19

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Xinmei Tan	Official release
1.1	2024-11-19	Xinmei Tan	Removed Android 13 from Section 1.2 MT8676 Basic Information

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1 MT8676 General Introduction

1.1 MT8676 Overview

The MT8676 is a high-end automotive-grade 5G System-On-Chip (SoC) launched by MediaTek. It is designed for integrated cockpit and parking systems, offering exceptional cost-performance.

The MT8676 device is a highly integrated and scalable application automotive processor with rich multimedia features and AI capabilities. The chip integrates 4 Arm® Makalu and 4 Arm® Klein, along with a powerful multi-standard video codec. Furthermore, an extensive set of interfaces is integrated for camera modules, external audio components, UFS/SD cards, and external modules.

The ARM® Makalu application processor offers substantial computing power to support the latest open-source operating systems, as well as In-Vehicle-Infotainment (IVI) and Cockpit Domain Controller (CDC) applications. The ARM® Klein provide adequate computing power for dedicated tasks, such as software-based graphic rendering and real-time operating system in critical execution environments.

In addition, the SMMU with unique performance enhancements for hard real-time masters, integrated in MT8676, provide a straightforward hardware mechanism that minimizes the virtualization overhead of hypervisors. This is particularly beneficial when multiple operating systems coexist for various application domains, thus optimizing overall system performance.

The new Gen 7 APU in MT8676 is capable to adapt to the latest AI trends in achieving maximizing effective performance in AI-multimedia, AI-camera, and AI-voice experiences. The APU Gen 7 also aims to ensure that AI-enhanced technologies perform sustainably across various conditions for extended durations.

The multi-standard video accelerator and an advanced audio subsystem are also integrated to provide advanced multimedia applications and services such as streaming audio and video, a multitude of decoders and encoders. The high-performance CPU, DSP, and hardware coprocessors are combined to provide a powerful modem subsystem capable of supporting NR Sub6, LTE Cat 18, Category 24 HSDPA downlink and Category 7 HSUPA uplink data rates, as well as Class 12 GPRS, EDGE.

MT8676 also embodies wireless communication devices, including WLAN, Bluetooth and GPS. With four advanced radio technologies by Combo chipsets, MT8676 provides the best and most convenient connectivity solution in the industry.

The chip supports rich automotive camera features, such as around view monitoring, back view monitoring, automotive driving recording, and driver monitoring. These features can be employed via the maximum 16x video stream inputs with MIPI-CS12, or Ethernet AVB inputs. The rich display interface (MIPI-DSI and DP) allows for support of up to 6 display panels, further enhancing its usability in various automotive applications.

1.2 MT8676 Basic Information

- **System:** Android U

- **Kernel:** kernel-6.1
- **Chipset:** MT8676 + MT6197 (RF) + MT6637 (CON) + MT6363 & MT6373 & MT63190 & MT6319U (PMIC)

1.3 MT8676 System Block Diagram

MT8676 system block diagram is shown in Figure 1-1:

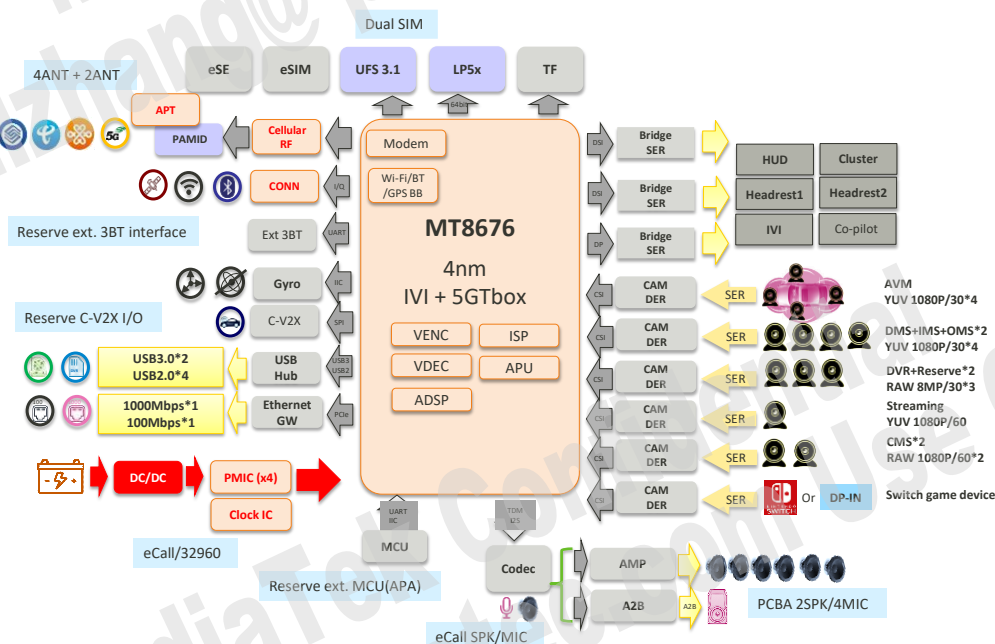


Figure 1-1. MT8676 system block

1.4 MT8676 SoC Specification

MT8676 SoC specification is shown in Table 1-1:

Table 1-1. MT8676 SoC specification

Category	5G SoC MT8676
Process	4nm
CPU	170K DMIPS
GPU	Mali-G615 MC6 1.8T FLOPS
APU	MVPUx1 + MDLAX2, 21.5TOPS
DRAM	LP5x 64bit/7500Mbps, 24GB
Flash	UFS4.0 2lane x1
Camera	19CAM, CSI x6
Display	6 Display DSI 8lane/2.5Gbps + DP1.4 4lane Up to 8K1K
Video Decoder	4K/60, H.264/H.265
Video Encoder	4K/60, H.264/H.265
Audio	ADSP HiFi3*2/800MHz TDM: 8OUT&8IN, I2S: 4OUT&5IN
Modem	5G 3CC R16 & LTE Cat-19
Connectivity	Wi-Fi 6E, BT5.3, GNSS 4M2B/10Hz
Ethernet	Ext PCIe MAC/PHY
PCIe	PCIe3.0 1lane 8Gbps x1
USB	USB3.2 Gen 1 x 1
OS	Android U
AECQ & ASIL	AECQ-104, QM/System cluster ASIL-B

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